

SESSION 1 — ACTIVITY (20 MIN) • STUDENT HANDOUT

Decode the Chart

Name: PRAHARSHINI Roll No: 13

For each chart below, answer: What is it trying to tell you? Do you trust it? Why or why not?

For each chart below, answer: What is it trying to tell you? Do you trust it? Why or why not?

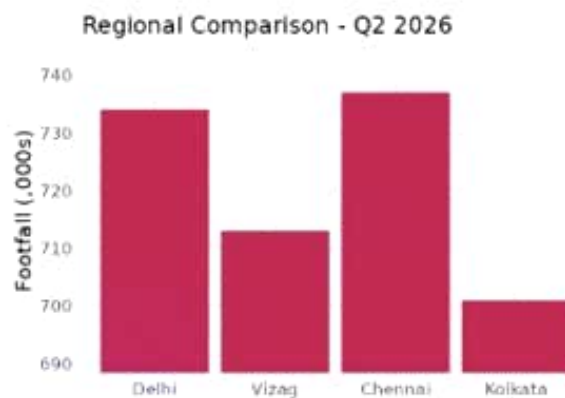


Chart 1

App Usage by Category (9 categories)



Chart 2

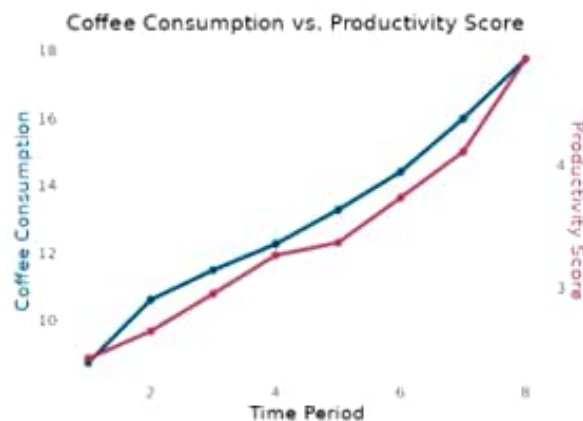


Chart 3

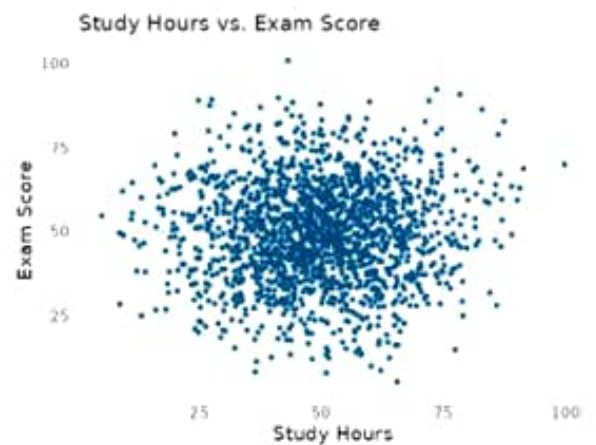


Chart 4

1-10-2026

Activity - Decode the chart

Chart 1 - Truncated Y-Axis

What is it trying to tell you?

It compares the values of different cities using a bar chart

Do you trust it? Why or why not?

No, not completely. The y-axis does not start from zero. This makes small differences between the bars look much larger than they actually are

Problem:- Truncated y-axis

Correct approach: A bar chart should normally start its y-axis at

Chart 2 - Too many Categories in pie chart

What is it trying to tell you?

It shows how app usage is divided among different categories

Do you trust it? Why or why not?

Not completely. There are too many slices, making it difficult to compare the categories accurately using colours and angles.

Problem: Too many categories

Correct approach: use a sorted bar chart when there are

more than about 6-8 categories.

Chart 3 - Dual y-Axis / False Correlation

It appears to show that coffee consumption and productivity increase together

No, Not Directly. The two variables use different y-axis. The scales can make the lines appear to move together, even when there may not be a real relationship

probm: Dual y-axis and possible false correlation

correct: use a common scale, separate charts, or calculate a correlation statistic

Chart 4 - Overplotting

- It shows the relationship between study hours and exam scores.
- Not completely. There are too many overlapping points, creating dense blobs. It is difficult to see where the data is actually concentrated.

probm overplotting

correct use transparency (α), a 2D histogram / hexbins or contour lines.